

in audio/video packetised elementary stream (PES) and data packets known as sections, as per DVB standards. PES is a mechanism to carry audio/video elementary streams in packet format inside an MPEG-2 transport stream. Sections are the data packets containing information regarding the audio/video content and other metadata. Once the demux does its job, audio/video is sent to MPEG-2/MPEG-4 decoder which gives the output to the RF port to display the video on television.

The data packets are sent to the processor used by the STB software to enable viewing. This data can be persistent or kept in RAM as per the needs and performance. Any kind of user request, be it tuning or a purchase, goes through the CPU. In between, the most important functionality of descrambling is performed by a descrambler embedded inside the ICAM part of the chipset. In some variants of STBs, a descrambler can be a part of a decoder, or it can exist independently. The descrambler takes care of decrypting the encrypted transport stream using the control word technique. Generally, these descramblers and control word algorithms are closely-guarded secrets with CA solution providers to prevent hackers from decrypting the signal. These are developed in conjunction with chipset vendors in a very secured environment.

Memory

Moving on further, let's discuss some of the memory types used inside an STB. It contains various memory types for different roles.

- **DRAM** (dynamic random access memory) is used to store all the software needs and volatile working area like static variables.
- **Video DRAM** is used in an STB for storing the decoded video image that is being output to television or video recorder.
- **NVRAM** (non volatile random access memory) is needed for persistent memory needs of the STB. Generally, STB configurations are written into NVRAM to keep the settings persistent

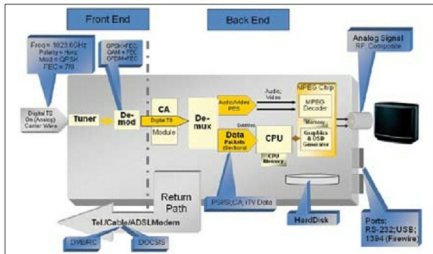


Fig. 3: Flow control between various STB hardware components

during power cycles and provide consistent user experience. There are two types of NVRAMs used inside a set-top box, namely, Electrically Erasable Programmable Read Only Memory (EEPROM) and FLASH. EEPROM is phasing out slowly as it is very expensive to perform frequent read and writes, whereas FLASH uses batch mode for read and write and extends the life of NVRAM. Also, FLASH can be used for storing parts of code segment for software performance reasons.

- **ROM/OTP** (read only memory/one time programmable): As the name suggests, this is the memory written on to once in a lifetime of set-top box. It contains some crucial software pieces like boot loader and other security-related code. The boot loader enables the STB to check the integrity of the software and let it work only if it contains authentic software. This is one of the key memory areas which protect the STB from hackers to write illegal software into the box.

As mentioned earlier, the most important part in an STB design is ICAM or CA module, which is responsible for the viewer's access to the content available on air. It maintains all the transactions and performs validations to check the entitlement of the viewer to selected programme. ICAM module consists of Entitlement Control Message (ECM) and Entitlement Manage-

ment Message (EMM) acquisition and management, descrambler and smart card interface. ECMs and EMMs are the ways of communication used by broadcaster to an STB. Smart card interface provides access to secured information like passwords, parental lock information, purchase information and all other kinds of transactional details. The most important part is the descrambler that uses the secure control word to descramble the video. It is the key to enable the video to be decoded. The input transport stream is sent to descrambler as an input having a specific parity. Depending on this parity, respective control word is generated and used to descramble the video. The descrambling is done in hardware due to performance and security reasons. Doing it in hardware does not expose the control word beyond silicon and it does it much faster without stopping the input stream.

In a nutshell, set-top box hardware architecture is very similar to architectures of any recorder and replay device. But its uniqueness lies in its security model and the enhanced interactivity with viewers, enabling them to perform various actions during live video and also pay only for the content they watch. ●

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